

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce

Department of Information Technology (Bsc.IT Semester III)

Python for Data Science Practical

Program - I

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Class :SYIT	Batch :
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A farmer wants a program to perform basic calculations related to farming.

The program should:

- Display the farmer's profile.
- Add the weights of two harvested crops.
- Find the square root of the total harvested area.
- Calculate the area of a triangular field.
- Swap the quantities of two fertilizer bags if entered in the wrong order.

Code:

```

import math
# 1. Display Farmer's Profile
name = "Ram"
age = 45
village = "Kanpur"

print("=== Farmer's Profile ===")
print("Name      :", name)
print("Age       :", age)
print("Village  :", village)

# 2. Add the weights of two harvested crops
crop1 = float(input("\nEnter weight of first harvested crop (kg): "))
crop2 = float(input("Enter weight of second harvested crop (kg): "))

total_weight = crop1 + crop2
print("Total harvested crop weight =", total_weight, "kg")

# 3. Find the square root of the total harvested area
area = float(input("\nEnter total harvested area: "))
print("Square root of harvested area =", math.sqrt(area))

# 4. Calculate the area of a triangular field
base = float(input("\nEnter base of the field: "))
height = float(input("Enter height of the field: "))

triangle_area = 0.5 * base * height
print("Area of triangular field =", triangle_area)

# 5. Swap the quantities of two fertilizer bags
bag1 = int(input("\nEnter quantity of first fertilizer bag: "))
bag2 = int(input("Enter quantity of second fertilizer bag: "))

print("Before swapping:")
print("Bag1 =", bag1, "Bag2 =", bag2)

# Swapping
bag1, bag2 = bag2, bag1

print("After swapping:")
print("Bag1 =", bag1, "Bag2 =", bag2)
print("Sakshi Patade S026")

```

Output:

=== Farmer's Profile ===

Name : Ram

Age : 45

Village : Kanpur

Enter weight of first harvested crop (kg): 45

Enter weight of second harvested crop (kg): 66

Total harvested crop weight = 111.0 kg

Enter total harvested area: 625

Square root of harvested area = 25.0

Enter base of the field: 20

Enter height of the field: 40

Area of triangular field = 400.0

Enter quantity of first fertilizer bag: 100

Enter quantity of second fertilizer bag: 300

Before swapping:

Bag1 = 100 Bag2 = 300

After swapping:

Bag1 = 300 Bag2 = 100

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